

AI-Based Thermoplastic Extrusion Quality Prediction Report

Date: 20-07-2026

Time: 04:00 PM

Prediction Summary

Prediction	FAIL
Failure Risk	69.18%
Risk Level	Moderate Risk
Likely Defect	OD Out of Tolerance
Confidence	97.29%

Input Parameters

Parameter	Value
Material	PU
Braid	None
Machine	EXT-01
Target OD (mm)	6.0
Target ID (mm)	2.0
Wall Thickness (mm)	2.0
OD Tolerance (mm)	0.1
ID Tolerance (mm)	0.1
Wall Tolerance (mm)	0.15

Vacuum (bar)	-0.4984
Screw Speed (RPM)	53.85
Zone 2 Temp (°C)	198.56
Zone 3 Temp (°C)	192.74
Melt Pressure (bar)	152.44
Puller Speed (m/min)	19.627
Material Moisture (%)	0.1167

AI Recommendations

- Increase Melt Pressure from 152.44 to 160.06 bar
- Decrease Puller Speed from 19.63 to 18.65 m/min
- Decrease Wall Thickness from 2.00 to 1.90 mm
- Decrease Vacuum from -0.50 to -0.52 bar
- Decrease Material Moisture from 0.12 to 0.11 %

Generated by: AI-Based Thermoplastic Extrusion Quality Prediction System